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Multi-Agent Debate Simulation

Can AI characters change each other's minds?

Topic: Should students use AI tools in education?
5 Trials · 5 Rounds each · 4 Characters

Alice — Optimistic	Bob — Skeptical	Carol — Mediator	David — Cautious

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How the Simulation Works

1 Set up characters

Four AI personas are created with different starting beliefs and goals about AI in education.

2 Run a round

Each character reads what others said, thinks about it, updates their own belief, and writes a new message.

3 Memory

Important moments are saved to each character's memory so they can reference them in later rounds.

4 Repeat × 5 rounds

The debate continues for 5 rounds per trial, giving characters time to shift or hold their ground.

5 Analyze

After all 5 trials finish, charts and summaries are generated to show what changed and why.

Each trial is fully independent — characters start fresh with no memory of previous trials.

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Meet the Characters

Alice

Optimistic

Starting belief: AI tools can support learning and creativity, but I'm still figuring out where the right limits are.

Goal: Promote the benefits of AI in education while staying open to criticism.

Bob

Skeptical

Starting belief: AI tools carry real risks of dependency and misuse, though I can imagine responsible use cases.

Goal: Surface the risks and blind spots in overly optimistic views.

Carol

Mediator

Starting belief: AI tools are useful, but schools need clear boundaries.

Goal: Mediate between both sides and push the group toward balanced guidelines.

David

Cautious Supporter

Starting belief: AI tools can help students, but only when used responsibly.

Goal: Support useful adoption of AI while emphasizing moderation.

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Chart 1 — Did Their Opinions Change Each Round?

Each line is one of the 5 debate trials. The Y-axis shows whether the character was supportive, balanced, or skeptical in that round. Overlapping lines mean the character behaved consistently no matter the trial.

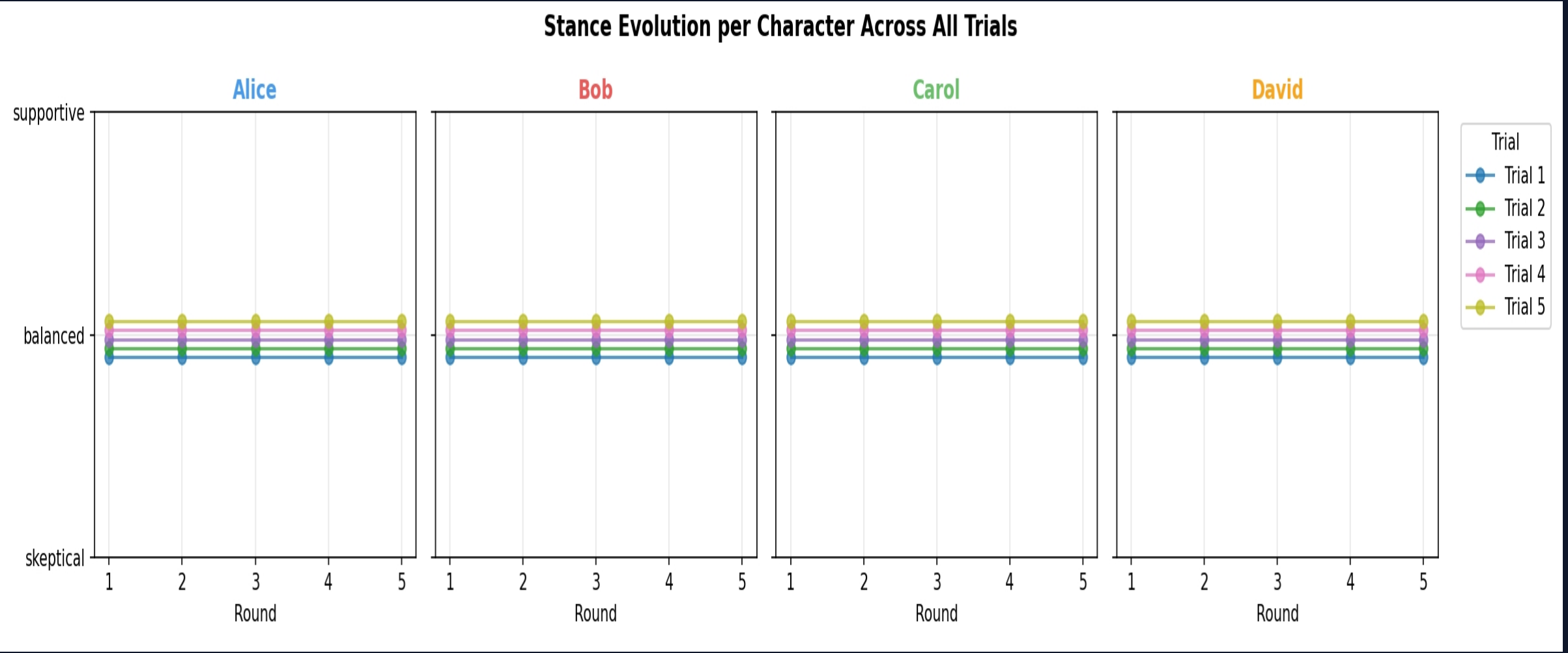

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Chart 2 — Did the Group Start Agreeing Over Time?

Lower values = the 4 characters' opinions are closer together. A downward curve means the debate brought them toward agreement. A flat or rising curve means they stayed divided.

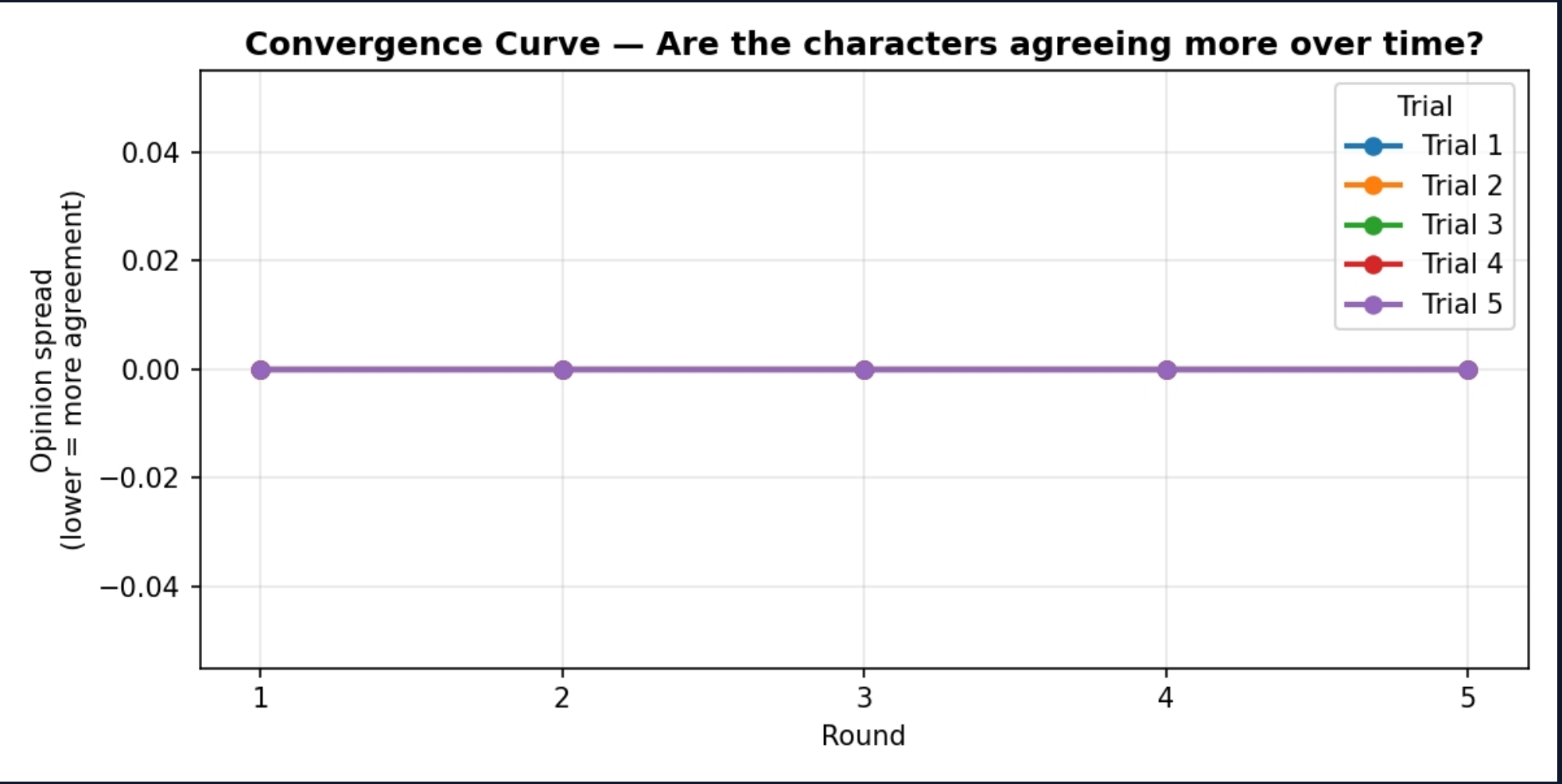

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Chart 3 — Who Was Convincing Whom?

Arrows point from the influencer to the person they moved. Thicker arrows = that influence happene
arrow is the total count

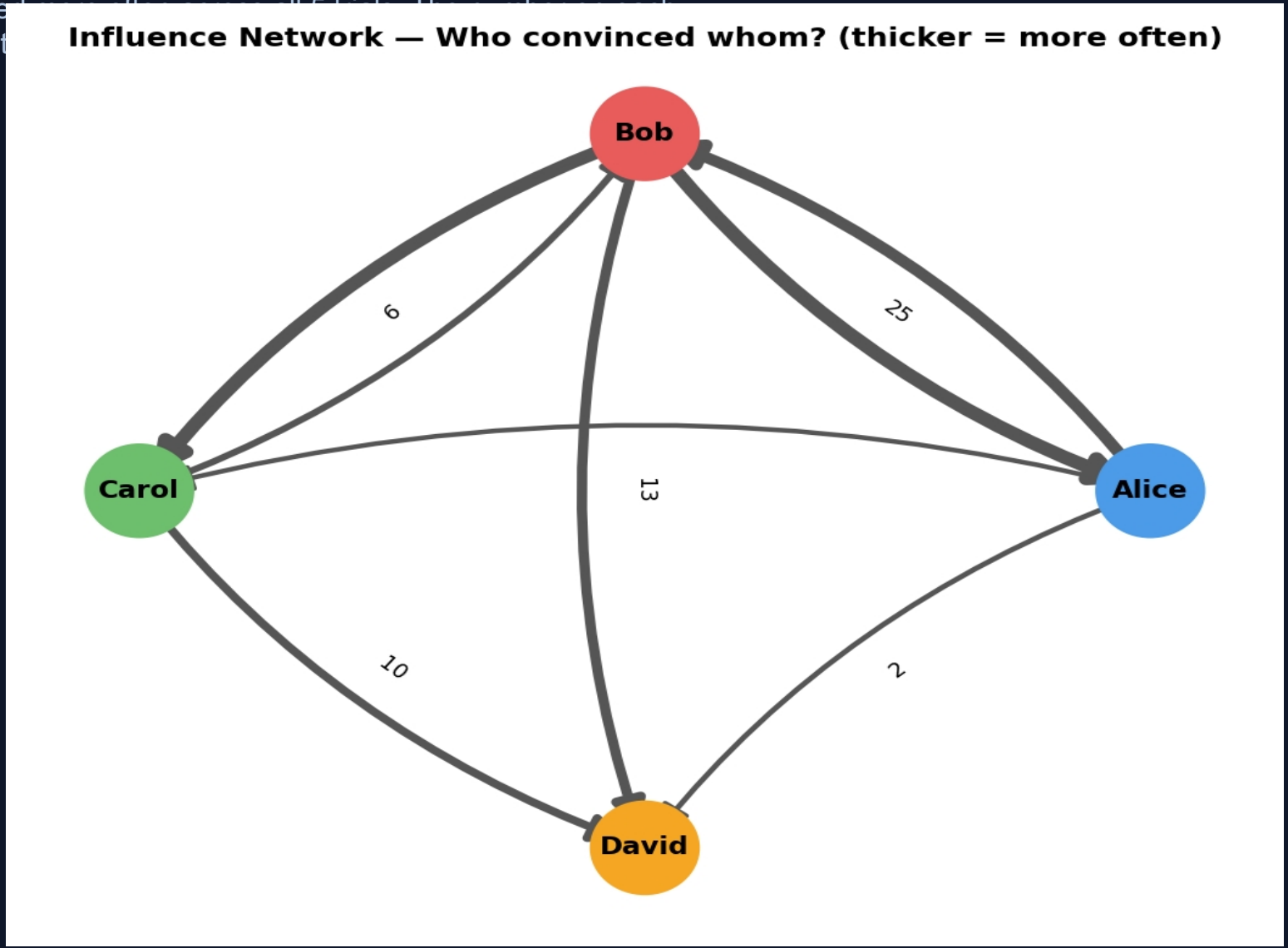

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Chart 4 — How Much Did Each Character Actually Change?

0 = no change from starting belief. 1 = completely shifted. The shaded band shows how consistent this was — a wide band means some trials moved them a lot, others barely at all.

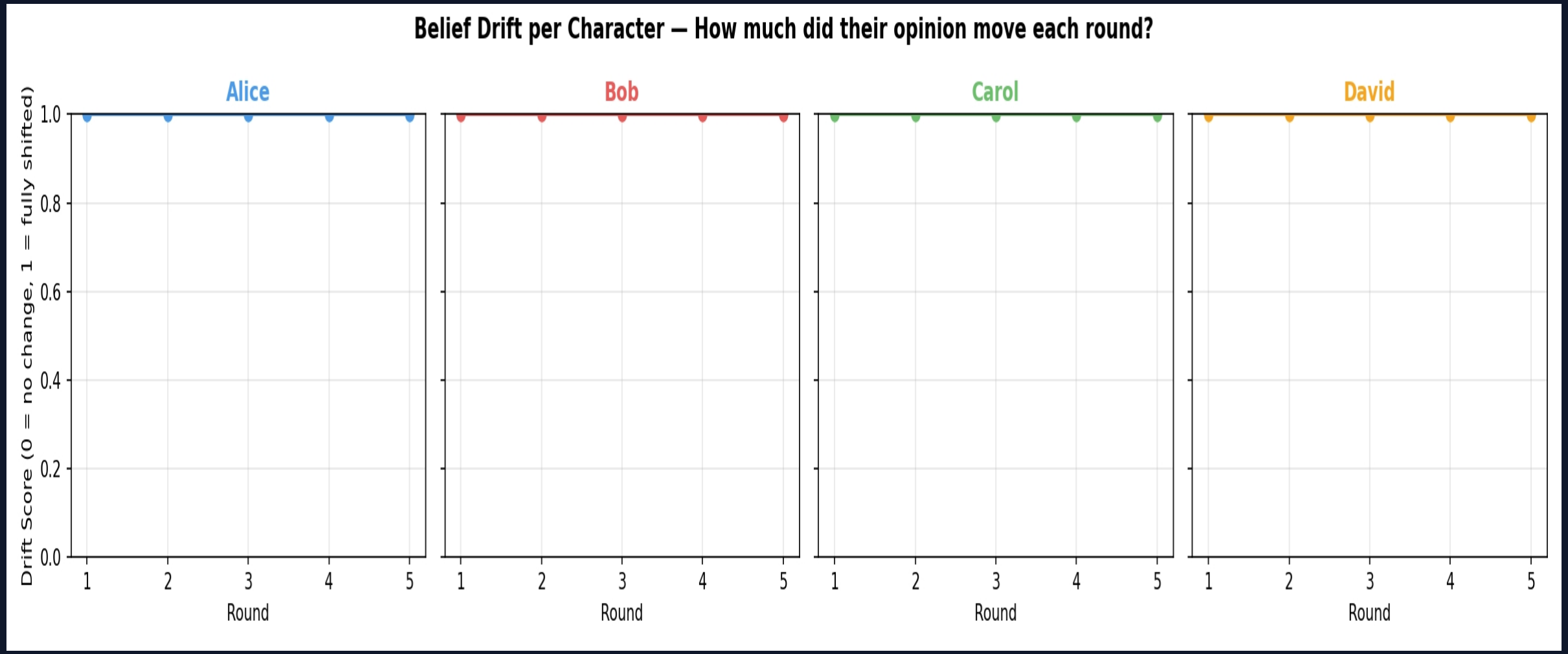

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Chart 5 — Which Topics Kept Coming Up?

Each circle pair compares two consecutive trials. The overlapping center = topics both sessions discussed. The outer edges = topics unique to just one trial.

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Key Findings

Broad agreement emerged

Across all 5 trials, every character ended up agreeing that AI in schools requires careful guidelines and a strong focus on preserving students' critical thinking.

Bob was the hardest to move

Bob (the skeptic) held his ground most consistently — his belief that AI poses real cognitive risks stayed firm in nearly every trial.

Carol and David drove convergence

Carol (mediator) and David (cautious supporter) were the characters most likely to shift toward common ground, helping pull the group together.

Critical thinking was universal

The theme of 'critical thinking' appeared in all 5 trials without exception — it was the single most consistent topic the simulation gravitated toward.

Over-reliance was the core fear

Even the supportive characters (Alice, Carol) ended up worried about students becoming too dependent on AI — the risk of over-reliance outlasted any initial optimism.

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Where Each Character Ended Up

Final beliefs from the last trial:

Alice

"AI tools must be integrated with structured activities that explicitly foster independent learning and critical evaluation, with a focus on preventing passive acceptance of AI-generated content."

Bob

"I now recognize the importance of actively guarding against students developing an unhealthy dependence on AI-generated responses."

Carol

"Schools must prioritize developing students' critical thinking skills alongside the use of AI tools."

David

"AI tools can assist students effectively only when designed to actively promote critical thinking skills and necessitate deliberate questioning, particularly regarding the potential for passive acceptance of AI responses."